

of seconds is still satisfactory.

7.19.2.5. **POSIX Time in milliseconds Type**

This type represents an offset, in milliseconds, from the UNIX epoch (1970-01-01 00:00:00 UTC), encoded as an unsigned 64-bit scalar value.

This type is employed for compatibility reasons.

7.19.2.6. **System Time in microseconds Type**

System time in microseconds is an unsigned 64-bit value representing the number of microseconds since boot.

7.19.2.7. **System Time in milliseconds Type**

System time in milliseconds is an unsigned 64-bit value representing the number of milliseconds since boot.

This type is employed for compatibility reasons.

7.19.2.8. **Elapsed Time in seconds Type**

Elapsed time in seconds is an unsigned 32-bit value representing the time that has elapsed for an operation or other activity, as determined by the definition of the attribute using this type.

7.19.2.9. **Temperature Type**

This type, derived from int16, represents a temperature on the Celsius scale with a resolution of 0.01°C.

- $value = (temperature\ in\ ^\circ C) \times 100$

- $-4^\circ C \Rightarrow -400$
- $123.45^\circ C \Rightarrow 12345$

The range is constrained by absolute zero: -273.15°C to 327.67°C.

Conversion of Temperature Values for Display

When converting temperature values for display manufacturers SHOULD ensure that calculations **round** to the nearest representable value. Particular care is needed when using integer arithmetic.

For example, assuming a display resolution of 0.5:

Attribute value	Temperature		Display as	
	°C	°F	°C	°F
1965	19.65	67.37	19.5	67.5